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Studierichting: Informatica
Oefeningenreeks 3E nr. 16

$$16.1 \ v(t) = 340(200e^{-200t})$$

$$\Rightarrow a(t) = 68000e^{-200t} km/u^2$$

$$\Rightarrow a_{max} = 68000 km/u^2$$

$$\Rightarrow 1 km/u^2 = \frac{1000}{3600^2} = 7.716 \cdot 10^{-5} m/s^2$$

$$\Rightarrow a_{max} = 68000 \cdot 7.716 \cdot 10^{-5}$$

$$\Rightarrow a_{max} \approx 5.25 m/s^2$$

$$\Rightarrow 0.7g = 0.7 \cdot 0.81 \Rightarrow 0.7g \approx 6.87 m/s^2$$

$$\Rightarrow 5.25 < 6.87 \text{ Dus de comfortnorm wordt gehaald}$$

$$16.2 \ F = ma \quad F = mg$$

$$m = 100 kg \quad a_{max} = 5.25 m/s^2 \quad g = 9.81$$

$$f_a = 525 N \quad f_g = 981 N$$

$$\Rightarrow F = \sqrt{981^2 + 525^2}$$

$$\Rightarrow f = \sqrt{1237986} \Rightarrow f \approx 1113 N$$

$$\Rightarrow 1 kg = 9.81 N \Rightarrow \frac{1113}{9.81} = 113.46 kg$$

References